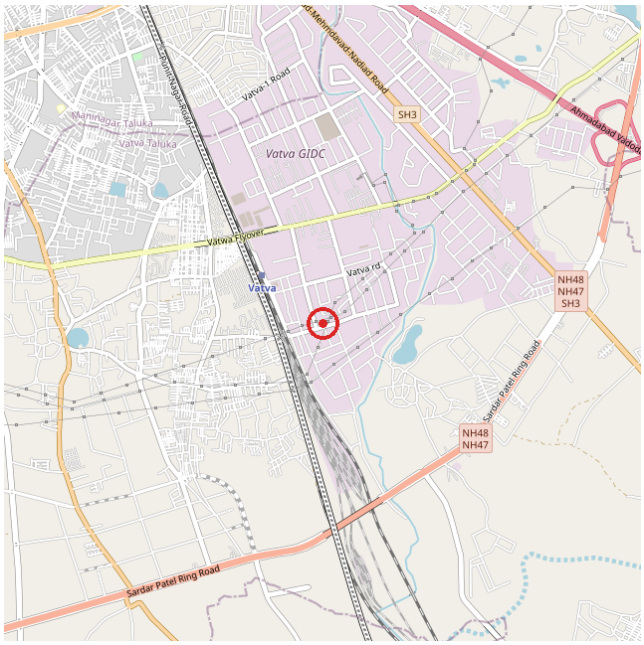
Incident report - alert #111 HIGH

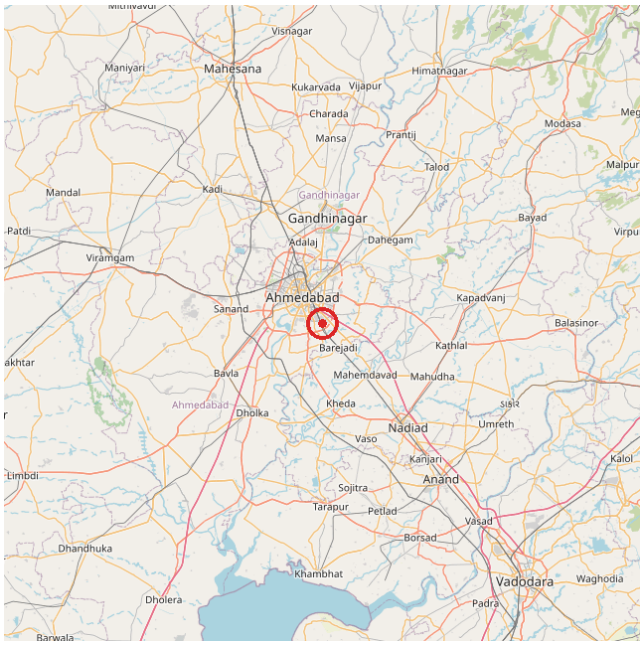
HIGH: possible industrial fire at JM Zhongkai Chemical Equipment LLP (factory generic), Gujarat - peak 62 MW; P(industrial fire) 95%; power +82.7 sd above this facility's own baseline

Place	JM Zhongkai Chemical Equipment LLP (factory generic), Gujarat
Location	22.95521 N, 72.63503 E
First detection	2020-12-08 20:03 UTC
Latest detection	2020-12-08 20:53 UTC
Peak power	62.4 MW at 2020-12-08 20:03 UTC
Why it alerted	both rules - P(fire) 95%, +82.7 sd vs own baseline
Status	Open

Where



Neighbourhood (zoom 14) - red ring = detection



Region (zoom 9) - red ring = detection

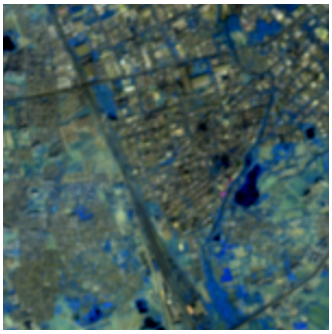
Map data (c) OpenStreetMap contributors (ODbL). Fire data: NASA FIRMS VIIRS, 375 m pixels.

What the satellite saw

Sentinel-2 L2A **S2A_42QZL_20201213_1_L2A**, 2020-12-13 05:52 UTC, 4% scene cloud, 0% over the chip; before: S2B_42QZL_20201208_1_L2A (2020-12-08 05:52 UTC).



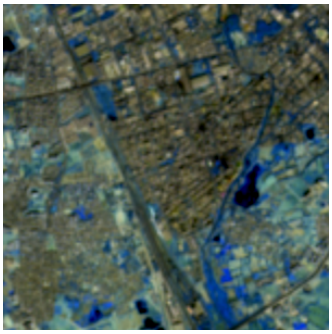
After - true colour



After - SWIR (flames glow orange)



Before - true colour



Before - SWIR

dNBR (95th percentile) +0.02 - unburned / no change; 0% of the 3 km chip changed like a burn scar. A fire inside buildings leaves no scar at 10-20 m, so a low value does not exclude one.

13 SWIR hot pixel(s) in the chip (B12 above B11: flames or very hot surfaces; furnaces and flares also qualify).

Landsat landsat-8 2020-12-08 05:33 UTC: surface temperature up to 313 K, 4 K above the chip median.

Reasons

• lies on a mapped factory generic (JM Zhongkai Chemical Equipment LLP)

• short-lived: only 1 day(s) of detections

• high radiative power (max 62 MW)

• active within the last 24 h

• this facility type is usually not thermally active, so a detection is notable

• Sentinel-2 2020-12-13 (4% scene cloud, 0% over the chip) vs 2020-12-08: dNBR p95 +0.02 (unburned / no change), 0% of the 3 x 3 km chip changed like a burn scar

• 13 SWIR hot pixel(s) in the 2020-12-13 scene (B12 flame / hot-slag signature)

• Landsat TIRS 2020-12-08: surface temperature up to 313 K, 4 K above the chip median

Timeline

2020-12-08 20:03 UTC	First detection of this source
2020-12-08 20:03 UTC	Peak power 62.4 MW
2020-12-08 20:53 UTC	Latest detection
2020-12-09 00:00 UTC	Alert raised (high, trigger both)
2020-12-13 05:52 UTC	Sentinel-2 S2A_42QZL_20201213_1_L2A, 4% cloud, dNBR p95 +0.02

Generated 2026-10-16 10:54 UTC by ThermoTrace. A thermal detection is evidence that a place was hot at that time, not a confirmed fire; the class and the alert come from heuristic rules and a model trained on rule-derived labels (see docs/EVALUATION.md). Verify with imagery and ground reports before acting. Data: NASA FIRMS (LANCE), OpenStreetMap contributors (ODbL), ESA WorldCover 2021, Copernicus Sentinel-2 / USGS Landsat.